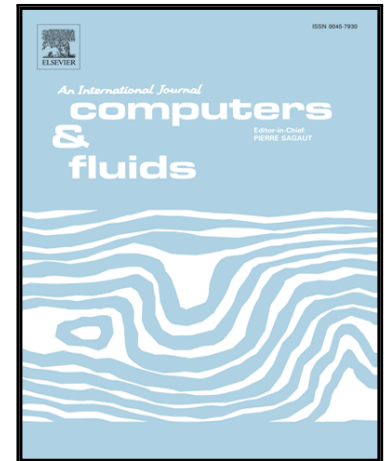


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Highlights

- We present a coupled ALE-AMR method for shock hydrodynamics on 3D unstructured grids
- Our method combines an unsplit ALE formulation with hierarchical refinement of tetrahedral cells
- Coupled ALE-AMR solutions are shown to converge to both uniform fine mesh solutions and analytic solutions Speed-ups over uniform fine mesh solutions of 5-10X for a given level of error are achieved

A coupled ALE-AMR method for shock hydrodynamics[☆]

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Abstract

We present a numerical method combining adaptive mesh refinement (AMR) with arbitrary Lagrangian-Eulerian (ALE) mesh motion for the simulation of shock hydrodynamics on unstructured grids. The primary goal of the coupled method is to use AMR to reduce numerical error in ALE simulations at reduced computational expense relative to uniform fine mesh calculations, in the same manner that AMR has been used in Eulerian simulations. We also identify deficiencies with ALE methods that AMR is able to mitigate, and discuss the unique coupling challenges. The coupled method is demonstrated using three-dimensional unstructured meshes of up to $O(10^7)$ tetrahedral cells. Convergence of ALE-AMR solutions towards both uniform fine mesh ALE results and analytic solutions is demonstrated. Speed-ups of $5 - 10\times$ for a given level of error are observed relative to uniform fine mesh calculations.

1. Introduction

Adaptive mesh refinement (AMR) [1, 2] and arbitrary Lagrangian-Eulerian (ALE) methods [3] both have long and successful histories in computational fluid dynamics, shock hydrodynamics, and related fields. AMR methods are typically implemented in an Eulerian framework [4, 5, 6] to increase mesh resolution at discontinuities, where solution accuracy is limited to first-order. ALE methods, which treat the mesh coordinates as an additional degree of freedom, are often implemented in a Lagrange-plus-remap form [7, 8, 9, 10] to maintain mesh robustness during extreme fluid distortion. ALE methods also can be used to allow mesh motion relative to the Eulerian reference frame for problems such as aeroelasticity and store separation [11, 12, 13, 14] as well as shock hydrodynamics [15, 16, 17].

Despite their individual records of success, few examples of coupled ALE-AMR methods can be found in the published literature. Anderson *et al* [18] presented a two-dimensional (2D) and three-dimensional

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(3D) structured grid approach with patch-based AMR that was applied to material deformation studies [19] and later implemented in a production ALE code [20]. Morrell *et al* [21, 22] presented a 2D structured grid approach based on cell-by-cell AMR. A more recent work by Baiges *et al* [23] used a coupled ALE-AMR method to compute incompressible free surface flows in both 2D and 3D. *hr*-adaptive methods [24, 25] are related but base mesh motion on error estimates rather than Lagrangian fluid motion or mesh quality concepts.

The primary benefit of a coupled ALE-AMR method is increased resolution at reduced computational expense relative to uniform fine mesh calculations. For example, both [18] and [21] reported speed-ups of up to $8 - 10\times$ with ALE-AMR over uniform fine mesh ALE calculations on structured meshes. In this sense AMR has the potential to reduce time to solution for a given level of error for ALE methods, just as it can do so for Eulerian methods.

Additionally, certain limitations specific to ALE methods can be potentially mitigated through the use of AMR. For example, ALE methods can create coarse mesh regions as a consequence of concentrating mesh points near flow features. Secondary flow phenomena can then transport through these coarse mesh regions with high numerical errors that could be reduced by AMR. This is shown in Figure 1 for a pseudo-2D (finite thickness) confined point blast: mesh moves away from the blast origin as the initial shock expands, and the mesh cells in this region are $3-4\times$ larger than their original size when a reflected shock later passes through.

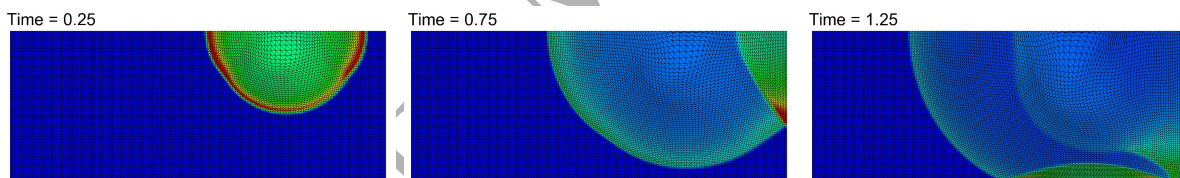


Figure 1: Mesh and pressure contours for a pseudo-2D confined point blast computed with an ALE method

ALE mesh motion based on the fluid velocity can also fail to track shocks when the sound speed is significantly faster than the fluid velocity, whereas AMR will increase resolution at shocks regardless of the wave speed for explicit schemes.

The goal of the current work is to develop a coupled ALE-AMR solution methodology for the simulation of unsteady compressible flows on 3D unstructured grids. Although such a methodology may appear to simply require that both ALE and AMR are implemented in the same code and compatible at the software level, this is a necessary but not sufficient condition. A useful analogy is coupled multi-physics simulation: different physics packages must not only be implemented in the same code but also coupled in a physically

correct manner. Similarly, ALE and AMR methods must be coupled in a *mathematically correct* manner. This coupling is therefore the primary subject of this paper.

The major challenges with a coupled ALE-AMR method primarily have to do with the design of mesh motion and mesh adaption schemes that are consistent with one another, rather than in conflict. Some specific issues include:

1. Interaction of ALE mesh motion strategies with AMR refinement boundaries
2. Preservation of mesh anisotropy introduced by AMR during ALE mesh motion
3. Conservation of cell volume during derefinement when child cells may no longer be aligned with their parent
4. Time step collapse due to simultaneous refinement (due to AMR) and compression (due to ALE mesh motion) of mesh cells

An important subtlety is that some of these issues are most effectively dealt with via *a priori* design choices in the individual ALE and AMR algorithms, rather than *a posteriori* modifications. The approaches used to overcome these issues constitute the novel aspects of this work relative to the previous studies cited above. Specifically, we extend and demonstrate the advantages of coupled ALE-AMR to high-speed compressible flows on 3D unstructured grids. The coupling of ALE and AMR is also what distinguishes this work from studies that have developed independent ALE and AMR variants of the same discretization scheme (*e.g.* [26] and [27]).

In what follows, we summarize the governing equations under consideration, and present the ALE and AMR algorithms. We then describe the mesh motion procedures for the coupled method. Lastly, the method is demonstrated on 3D test problems with mesh sizes of up to $O(10^7)$ cells.

2. ALE solution algorithm

In this work we are concerned with numerical solution of the compressible Euler equations, written here in the flux-conservative ALE form

$$\frac{1}{V} \frac{\partial}{\partial t} \begin{pmatrix} V\rho \\ V\rho u_i \\ V\rho E \end{pmatrix}_w + \frac{\partial}{\partial x_j} \begin{bmatrix} \rho(u_j - w_j) \\ \rho u_i(u_j - w_j) + \delta_{ij}p \\ \rho E(u_j - w_j) + p u_j \end{bmatrix} = 0 \quad (1)$$

where ρ is density, u_i is fluid velocity, w_i is the velocity of the reference frame, E is specific total energy, p is pressure, V is volume, and t is time. The subscript w on the time derivative is used to indicate that the

equations are written in a reference frame moving at some arbitrary velocity w_i relative to the laboratory frame. The Lagrangian and Eulerian limits are $w_i = u_i$ and $w_i = 0$, respectively. A generic equation of state of the form $p = p(\rho, e)$ is assumed (an ideal gas equation of state is used for test problems in what follows).

In what follows we use the condensed notation

$$\frac{1}{V} \frac{\partial (\hat{U}V)}{\partial t} + \frac{\partial \hat{F}_j}{\partial x_j} = 0 \quad (2)$$

where

$$\hat{U} = \begin{pmatrix} \rho \\ \rho u_i \\ \rho E \end{pmatrix} \quad (3)$$

is the vector of conserved unknowns and

$$\hat{F}_j = \begin{pmatrix} \rho(u_j - w_j) \\ \rho u_i(u_j - w_j) + \delta_{ij}p \\ \rho E(u_j - w_j) + \rho u_j \end{pmatrix} \quad (4)$$

is the flux; we also drop the subscript w from the time derivative for brevity of notation.

The system (2) is discretized with an edge-based Finite Element (FE) method for linear tetrahedra using node-centered unknowns. The continuous system is multiplied by a test function $N(x_i)$ and integrated over the domain Ω to give the weak form

$$\int_{\Omega} N(x_i) \left[\frac{1}{V} \frac{\partial (\hat{U}V)}{\partial t} + \frac{\partial \hat{F}_j}{\partial x_j} \right] d\Omega = 0 \quad (5)$$

Assuming a discretization of Ω into linear tetrahedral elements Ω_h and a nodal Galerkin approximation for the unknowns and fluxes, the semi-discrete system becomes

$$\sum_{\Omega_h \in v} \sum_{w \in \Omega_h} \int_{\Omega_h} N^v N^w \left[\frac{1}{V} \frac{\partial (\hat{U}V)}{\partial t} \right]^w d\Omega + \sum_{\Omega_h \in v} \sum_{w \in \Omega_h} \int_{\Omega_h} N^v \frac{\partial N^w}{\partial x_j} \hat{F}_j^w d\Omega_h = 0 \quad (6)$$

where v and w are nodal indices, N is a linear basis function on Ω_h , and the dependence on x_i has been dropped for brevity of notation. Following [15], a lumped mass approximation is used and the flux term is written in an edge-based form to give

$$\frac{d(\hat{U}V)^v}{dt} + \sum_{vw \in v} D_j^{vw} \hat{F}_j^{vw} = 0 \quad (7)$$

where now d/dt denotes the discrete time derivative, vw is an edge containing v , D_j^{vw} is a dual-mesh area vector given by

$$D_j^{vw} = \frac{1}{2} \sum_{\Omega_h \in vw} \left(N^v \frac{\partial N^w}{\partial x_j} - N^w \frac{\partial N^v}{\partial x_j} \right) d\Omega_h \quad (8)$$

and the summation in (7) is over edges connected to node v . A complete derivation of the edge-based form is given in [15] and therefore not repeated here. In addition to the Euler equations, the coordinate equation

$$\frac{dx_i}{dt} = w_i \quad (9)$$

also must be solved to account for the moving ALE reference frame, where in the discrete formulation w_i now represents the mesh velocity.

The numerical flux, $\bar{F}_j^{vw}$ in (7), is evaluated at the midpoint of each edge via the solution to a local Riemann problem. In this work the Rusanov scheme [28]

$$D_j^{vw} \hat{F}_j^{vw} = D_j^{vw} (\hat{F}^w + \hat{F}^v) - |D_j^{vw}| \lambda^{vw} (\hat{U}^w - \hat{U}^v) \quad (10)$$

is used where λ^{vw} is a wave speed defined on vw . Following [15] this wave speed is computed as

$$\lambda^{vw} = \max \left[\left| (u_j^v - w_j^v) \frac{D_j^{vw}}{|D_j^{vw}|} \right| + c^v, \left| (u_j^w - w_j^w) \frac{D_j^{vw}}{|D_j^{vw}|} \right| + c^w \right] \quad (11)$$

Other Riemann solvers, including the HLL and HLLC schemes [29, 30] as well as a multi-directional method [16, 31], have been used with the above spatial discretization. The Rusanov scheme is the focus for this work due to its robustness on non-uniform unstructured grids, particularly with strongly-shocked complex flows.

A MUSCL reconstruction with Van Leer limiting on primitive fluid variables is used to improve accuracy. For an edge vw with left and right variables f^v and f^w , a MUSCL reconstruction [32] for the edge mid-point values is written as

$$\tilde{f}^v = f^v + \frac{1}{4} \left[(1 - \mu) \hat{\phi}_L \delta_1 + (1 + \mu) \hat{\phi}'_L \delta_2 \right] \quad (12)$$

$$\tilde{f}^w = f^w - \frac{1}{4} \left[(1 - \mu) \hat{\phi}_R \delta_3 + (1 + \mu) \hat{\phi}'_R \delta_2 \right] \quad (13)$$

where ϕ is the Van Leer limiter function and μ is a constant. The quantities $\delta_{[1-3]}$ are defined as

$$\delta_2 = f^w - f^v \quad (14)$$

$$\delta_1 = 2 \frac{\partial f^v}{\partial x_i} x_i^{wv} - \delta_2 \quad (15)$$

$$\delta_3 = 2 \frac{\partial f^w}{\partial x_i} x_i^{wv} - \delta_2 \quad (16)$$

with $x_i^{wv} = x_i^w - x_i^v$. The resultant scheme is second-order accurate on smooth flows and first-order accurate at discontinuities. The complete method including 3D implementation details and verification studies is described in [15].

The semi-discrete system (7) is integrated in time with an explicit multi-stage scheme of the form

$$(\hat{U}V)^{v,k} = (\hat{U}V)^{v,0} - \alpha^k \Delta t \left(\sum_{vw \in v} D_j^{vw} \hat{F}_j^{vw} \right)^{v,k-1} \quad (17)$$

where $1 < k < m$ is the current stage, m is the total number of stages, Δt is the time step, and

$$\alpha^k = \frac{1}{1 + m - k} \quad (18)$$

For this work, $m = 2$ which corresponds to a second-order accurate Runge-Kutta scheme. The time step at each edge is calculated as

$$\Delta t^{vw} = \frac{c_N l^{vw}}{\lambda^{vw}} \quad (19)$$

where l^{vw} is the length of the edge, λ^{vw} is the wave speed on the edge computed from (11), and $c_N \leq 1$ is the Courant number. The global time step is the minimum of the Δt^{vw} . An additional time step control based on the volume change at node v is calculated as

$$\Delta t^{v,n} = c_V \Delta t^{n-1} \frac{\min(V^{v,n-1}, V^{v,n})}{|V^{v,n-1} - V^{v,n}|} \quad (20)$$

where c_V is a constant typically set to $0.01 - 0.05$. The coordinate equation (9) is integrated with the same scheme. Note that this is an unsplit or direct ALE method [35] with no explicit Lagrange step *i.e.* the governing equations are solved directly in the co-moving frame defined by w_i . The determination of w_i is discussed subsequently.

High-order unstructured finite volume methods, including high-order time integration, have been devel-

oped as an alternative to the above discretization for both Eulerian [33] and ALE [17] formulations. However, on problems dominated by shocks and material discontinuities, the high-order methods have been found to be at best competitive with – in terms of runtime for a given error – standard schemes. The scope of this work is therefore limited to typical second-order accurate Total Variation Diminishing methods with similar temporal accuracy.

3. Adaptive mesh refinement algorithm

The AMR algorithm used in this work is based on previously developed unstructured mesh concepts [6, 36, 37] and involves the hierarchical refinement of individual tetrahedral cells into two, four, or eight child cells. An error estimate (to be defined subsequently) is calculated on each edge, and a tetrahedral cell is flagged for refinement based on which of its associated edges satisfy some refinement threshold. Transition patterns also are included which allow cells refined into two or four child cells to refine into eight child cells before refinement to additional levels. The derefinement patterns are the inverse of the refinement cases, with the addition of an allowable derefinement from four to two child cells.

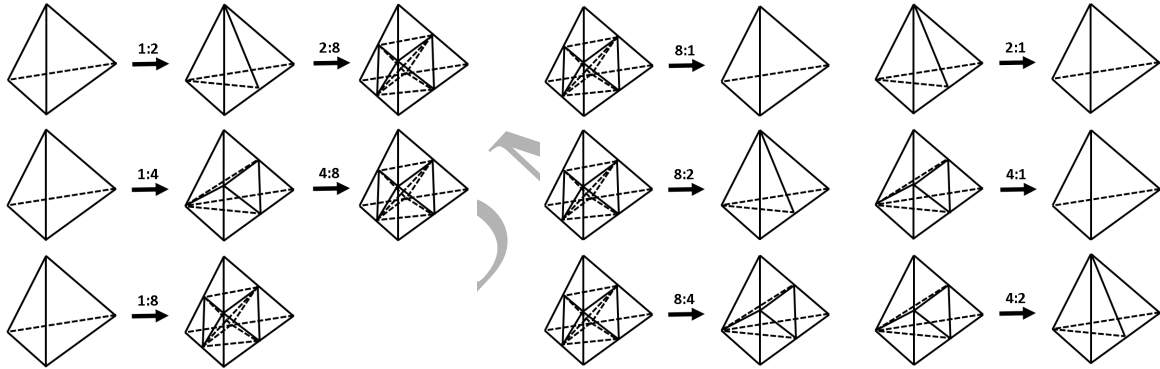


Figure 2: Allowable refinement (left) and derefinement (right) patterns

The refinement and derefinement patterns are illustrated graphically in Figure 2. A tree-like data structure is used to store the refinement history (including parent-child cell relations) and facilitate derefinement.

Although the refinement and derefinement patterns shown in Figure 2 include hanging nodes, an iterative procedure is used to eliminate hanging nodes from the refined mesh through refinement of neighboring cells. The iterative procedure itself is somewhat complicated (see [6] for details) in 3D, but is illustrated for a simple 2D example in Figure 3. The elimination of hanging nodes leads to a significant simplification for all other algorithms: since the refined mesh consists entirely of standard linear tetrahedra, the discrete operators do not need to be modified. The elimination of hanging nodes also results in a smooth transition from unrefined

to fully refined regions, as opposed to a distinct boundary between unrefined and refined regions. As a side effect, the iterative procedure has a tendency to diffuse the refinement region slightly. This behavior helps to ensure that mesh cells ahead of a flow feature refine prior to the arrival of the flow feature. The adaptive mesh procedure can therefore be applied at a frequency of approximately $1/c_N$ rather than every time step.

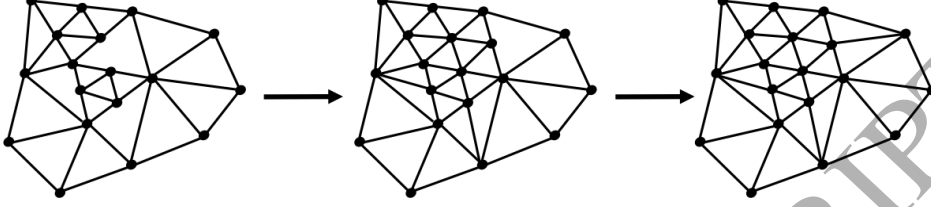


Figure 3: Elimination of hanging nodes through iterative refinement of neighboring cells: given the initial refinement pattern (left), neighboring cells are successively refined (middle and right) as needed until all hanging nodes have been eliminated

The error estimate is based on a normalized gradient. For some solution variable f , the error estimate on an edge vw is computed as

$$\epsilon_h^{vw} \propto h \left| \frac{\partial f}{\partial x_i} \right|^{vw} \approx \frac{|f^w - f^v|}{|f^w| + |f^v|} \quad (21)$$

This form is dimensionless and bounded on $[0, 1]$, which allows the use of multiple variables simultaneously. An edge is identified for refinement when ϵ_h^{vw} exceeds some threshold, typically $0.1-0.2$; and for derefinement when ϵ_h^{vw} drops below some threshold, which is typically half to a fourth of the refinement threshold.

4. Coupled mesh motion

A coupled ALE-AMR algorithm presents unique challenges for the mesh motion procedure. First, the mesh motion procedure must respect the mesh anisotropy introduced by the AMR algorithm. For example, mesh motion algorithms that attempt to equalize cell volume or edge length (see *e.g.* [38] for examples), or spring models (*e.g.* [39, 40]), will attempt to eliminate variations in mesh size, including those introduced by AMR, unless the algorithm is modified to explicitly account for such variations. A second related issue is stability of the mesh motion algorithm with respect to changes in mesh topology: since frequent changes in mesh velocity can lead to poor mesh quality and/or mesh tangling, the dynamics of the mesh velocity field should not be tightly coupled to the dynamics of the mesh topology. Lastly, because child cells must be aligned with their parent during derefinement to preserve volume, either the motion of child cells cannot be completely arbitrary, or the derefinement procedure must somehow account for misalignment between a parent cell and the union of its child cells. In this work the mesh motion procedure does not need

modification to account for hanging nodes, because hanging nodes are eliminated during the refinement procedure. Obviously, such modifications will be required if hanging nodes are present, as is typically the case in quadrilateral/hexahedral-based meshes.

A coupled mesh motion procedure that addresses these issues can be devised through consideration of the Geometric Conservation Law (GCL). The GCL is the governing constraint for moving mesh solution algorithms [41, 42, 43] and states that a uniform field must be invariant under arbitrary mesh motion. The discrete form of the GCL in the edge-based FE formulation is

$$\frac{dV^v}{dt} = \frac{1}{2} \sum_{vw \in v} D_i^{vw} (w_i^w + w_i^v) \quad (22)$$

In the Eulerian-AMR case, $w_i = 0 \forall v, t$ and the GCL simply implies that volume must be conserved during refinement and derefinement. In the ALE case, the GCL implies that the basis functions N^v and volumes V^v at time level $n+1$ must be computed consistently from the x_i^v at time level $n+1$. A coupled ALE-AMR method must satisfy both conditions

A mesh motion procedure that respects the mesh anisotropy introduced by AMR is achieved in this work through the requirement that a given mesh velocity field must be invariant under arbitrary mesh refinement. This requirement is directly analogous to the GCL statement that a uniform field be invariant under arbitrary mesh motion, and can be achieved through a physics-based mesh velocity, rather than a geometry-based relaxation technique. Since the goal of mesh motion in shock hydrodynamics applications is ultimately to track fluid motion – *i.e.* maintain a Lagrangian character to the simulation – the mesh velocity w_i is initially set equal to the fluid velocity u_i and approximated with the same linear basis functions. Additional control over w_i is provided through parameters that both scale the magnitude and filter out vorticity. The complete form is

$$w_i = \kappa u_i \quad (23)$$

with κ given by

$$\kappa = c_1 \max \left(0, 1 - c_2 \frac{|\omega_i|}{\|\omega_i\|_\infty} \right) \quad (24)$$

In the above expression, c_1 and c_2 are constants, ω_i is the fluid vorticity vector, and $\|\omega_i\|_\infty$ is the maximum-norm of ω_i .

The second step in the procedure is to smooth the mesh velocity with a discrete Laplacian operator to

minimize the likelihood of mesh tangling. The smoothing procedure involves solution of the discrete system

$$\sum_{\Omega_h \in \mathcal{V}} \int_{\Omega_h} N^v \frac{\partial^2 N^w}{\partial x_i^2} w_j^v d\Omega_h = - \sum_{\Omega_h \in \mathcal{V}} \int_{\Omega_h} \frac{\partial N^v}{\partial x_i} \frac{\partial N^w}{\partial x_i} w_j^v d\Omega_h + \sum_{\Gamma_h \in \mathcal{V}} \int_{\Gamma_h} N^v \frac{\partial N^w}{\partial x_i} n_i w_j^v d\Gamma = 0 \quad (25)$$

to some tolerance ϵ_w with a Conjugate Gradient method, where $\Gamma = \partial\Omega$ and n_i is the normal vector on Γ . Note that $\epsilon_w = 0$ will produce the Eulerian limit if $w_i = 0$ on the boundary, and a linear field if $w_i \neq 0$ on a subset of the boundary.

The parameter c_1 in (24) allows the overall simulation to be more Eulerian ($c_1 = 0$) or more Lagrangian ($c_1 = 1$). In principle this degree of control can also be accomplished by decreasing or increasing ϵ_w . However, the value of ϵ_w has a much larger impact on computational expense since it directly affects the number of smoothing iterations required. Therefore, decreasing c_1 is generally more cost effective than decreasing ϵ_w . The parameter c_2 allows regions of high vorticity to be treated in a more Eulerian-like manner, and regions of low vorticity to be treated in a more Lagrangian-like manner. Typical settings are $0.5 < c_1 < 1.0$, $c_2 = 1$, and $0.1 < \epsilon_w < 0.25$.

Because the mesh velocity is based on the local fluid velocity, the mesh velocity will be independent of the local mesh size to within truncation error. The mesh motion procedure therefore respects mesh anisotropy by construction. Although a mesh velocity approach does allow increased mesh irregularity to occur, the discrete operator used to solve (2) maintains second-order accuracy on distorted grids [15, 44].

An approach previously used in Lagrange-plus-remap ALE-AMR methods for volume conservation is to introduce an intermediate realignment and remapping step prior to derefinement [18]. This approach allows unrestricted Lagrangian motion of refinement points, but the additional remapping or advection step introduces both additional cost and additional numerical diffusion. Another alternative would be a generalized intersection-based remapping procedure [8] as employed in [23]. This approach is arguably the most general but also comes at increased computational expense.

In an unsplit ALE formulation with no Lagrange step, however, the motion of refinement points can be chosen such that alignment between child and parent cells is maintained through the choice of mesh velocity. In other words, the GCL constraints for both derefinement and mesh motion can be satisfied simultaneously through an appropriate choice of w_i for refinement points. For linear basis functions, the mesh velocity at a refinement point v' must be computed as

$$w_i^{v'} = \sum_{v \in \Omega_h} N^v w_i^v \quad \forall t \quad (26)$$

where Ω_h is the unrefined mesh cell that contains v . This follows from the fundamental relation

$$x_i = \sum_{v \in \Omega_h} N^v x_i^v \quad (27)$$

which expresses any spatial location x_i in terms of the barycentric coordinates defined on the host cell Ω_h . For any field variable f this expression also defines a generic linear prolongation operator that is consistent with both the underlying FE approximation and the time-dependent geometry. Note that this constraint cannot be used in a Lagrange-plus-remap method because it would unphysically restrict mesh motion during the Lagrangian step and suppress the additional degrees of freedom introduced by AMR.

The complete procedure for the specification of the mesh velocity is illustrated graphically in Figure 4 over a unit one-dimensional (1D) domain with mesh spacing of 0.25, and one level of mesh refinement such that refinement points are located at intervals of 0.125. The first curve is a notional non-linear Lagrangian mesh velocity shown by the solid line. This distribution corresponds to (23) with $c_1 = 1$ (c_2 is arbitrary in this example since the flow is 1D). The dashed line corresponds to the mesh velocity field after smoothing from a single Gauss-Siedel iteration on (25). Both the overall magnitude and the steepness of the profile have been reduced. Finally, the dotted line corresponds to the mesh velocity following the application of (26): the mesh velocity at refinement points has been set to the average of the parent values.

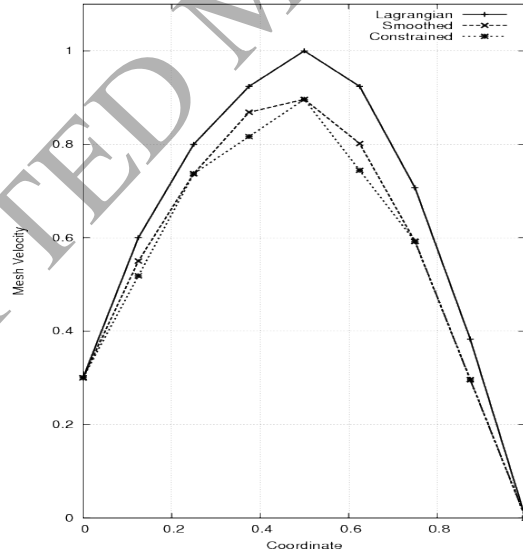


Figure 4: Lagrangian, smoothed, and constrained mesh velocities for a notional one-dimensional domain

The effect of (26) is further illustrated graphically for a two-dimensional patch of triangular cells in Figure 5. On the left, a patch of triangular cells experiences a notional deformation. The shaded region represents

a triangular cell that has been refined into three child cells, and adjacent triangles have been refined into two child cells to preserve a conformal mesh. Because the motion of refinement points is constrained, the child cells remain aligned with the original parent during mesh motion. On the right, however, the motion of refinement points is not constrained and the child cells do not remain aligned with the original parent. The deformed configuration on the left will therefore preserve volume during derefinement, while the deformed configuration on the right will not.

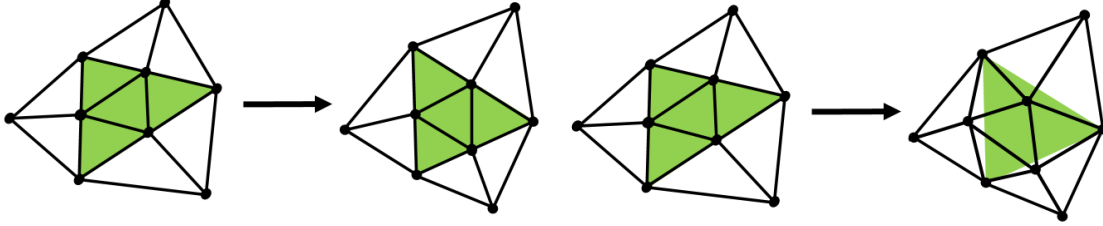


Figure 5: Constrained (left) and unconstrained (right) motion of child cells for a notional patch of triangular cells

The impact of (26) on the numerical solution is a change in advected volume given by

$$\Delta V = \Delta t \sum_{vw \in v} D_j^{vw} (\delta w_j^v + \delta w_j^w) \quad (28)$$

where δw_j is the difference between the smoothed and constrained velocity fields. This flux volume increment corresponds exactly to the volume difference between the original parent and the union of child cells shown in the rightmost mesh patch of Figure 5. The unsplit ALE formulation accounts for this difference automatically since the solution is calculated in the reference frame of the moving adapted grid.

5. Test Problems

5.1. Shock Tube

The first test case evaluated is a classic shock tube problem [45]. The computational domain is a square tube 100 units in length with unit width and height, and the initial pressure and density ratio across the interface is 10. An ideal gas equation of state with $\gamma = 5/3$ was used. Figure 6 shows the final density field and adapted surface mesh for ALE, one-level ALE-AMR, and two-level ALE-AMR simulations. Density was used as the error indicator variable. Figure 7 shows a close up of the adapted surface mesh, colored by mesh velocity, near the shock wave at intermediate and final times for the two-level ALE-AMR simulation. The conformal mesh and the smoothly varying mesh velocity field can both be observed.

Figure 8 shows density line plots through the center of the volume mesh for the two ALE-AMR cases, along with similar plots for one-level and two-level uniformly refined meshes. The ALE-AMR simulations

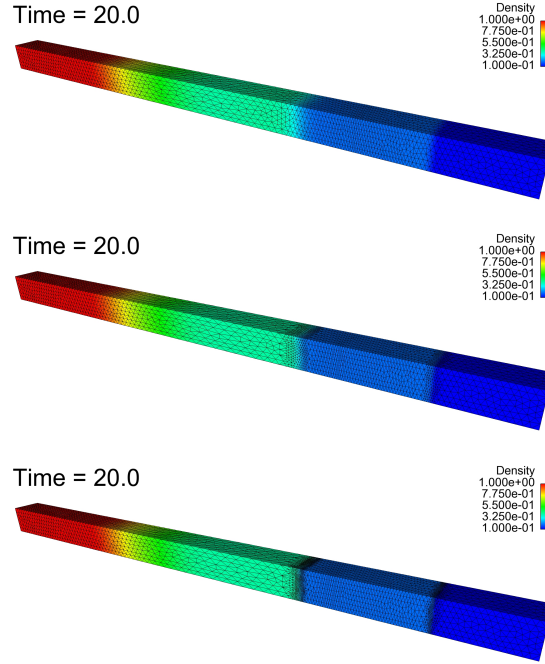


Figure 6: Final adapted surface mesh and density contours for a shock tube problem with ALE only (top), one-level ALE-AMR (middle), and two-level ALE-AMR (bottom)

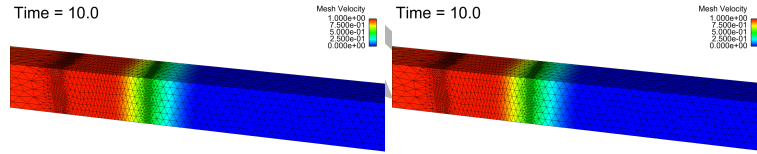


Figure 7: Adapted surface mesh and mesh velocity contours for the two-level ALE-AMR simulation at intermediate (left) and final (right) simulation times

agree well with the uniformly refined variants, which indicates convergence of the AMR method toward fine mesh simulation results.

Table 1 lists the L_1 density error, number of cells, number of time steps, and wall clock time for each variant using eight parallel threads. For the AMR calculations, the number of cells listed is the maximum observed during the simulation. Although the use of AMR clearly reduces numerical error, the convergence rate is slightly less than first-order due to the fact that the mesh is not refined in regions where the solution is smooth but slightly non-linear (such as the expansion region). This results in a higher error than would be achieved through uniform mesh refinement. The ALE-AMR calculations have an error approximately 9% higher for the one-level case and 27% higher for the two-level case. The slightly higher error is due to a

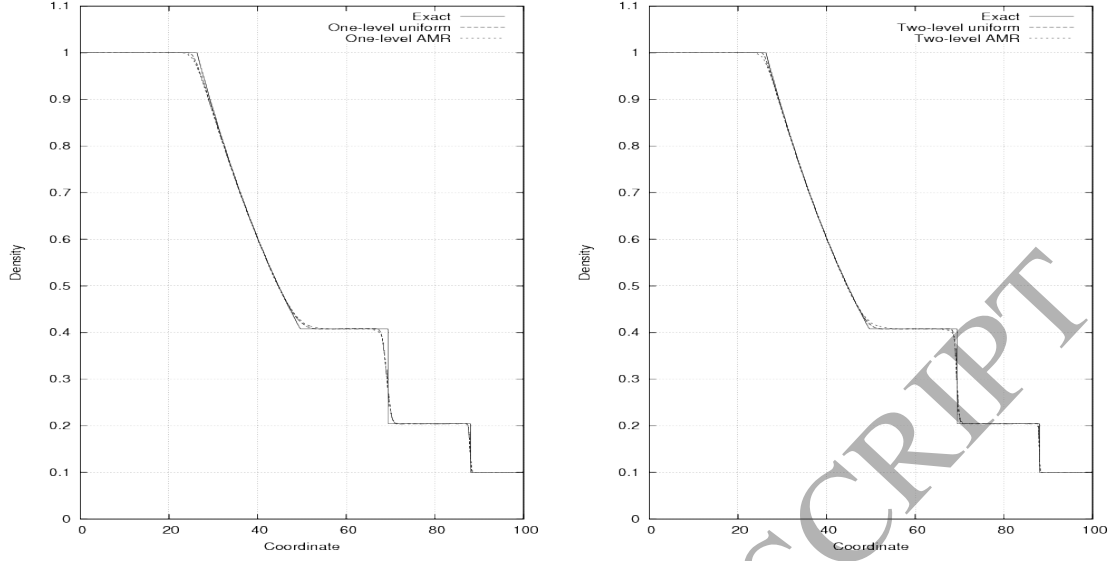


Figure 8: Density line plot for one-level ALE-AMR and one-level fine mesh ALE (left), and two-level ALE-AMR and two-level fine mesh ALE (right) simulations of the shock tube problem, including exact solution

Calculation	L_1 error	Number of cells	Runtime (sec)
ALE-only	6.02×10^{-3}	3.9×10^4	8
One-level ALE-AMR	3.51×10^{-3}	7.1×10^4	69
One-level fine mesh ALE	3.21×10^{-3}	3.1×10^5	147
Two-level ALE-AMR	2.27×10^{-3}	2.0×10^5	336
Two-level fine mesh ALE	1.78×10^{-3}	2.5×10^6	3089

Table 1: L_1 density errors, mesh sizes, and runtime information for ALE-only and ALE-AMR simulations of the shock tube problem

lack of refinement at the boundaries of the expansion region, as can be seen in Figure 6, and the associated increase in numerical diffusion, as can be seen in Figure 8. Nonetheless, the use of AMR provides a savings in cell count of approximately $4.4\times$ for the one-level case and $12.5\times$ in the two-level case. This translates to a runtime savings of approximately $2.1\times$ for the one-level case and $9.2\times$ for the two-level case. The efficiency gains are illustrated graphically in Figure 9, which shows error as a function of the degrees of freedom (*i.e.* number of mesh points). The ALE-AMR calculations clearly require less degrees of freedom to achieve a given level of error. The relatively low runtime savings for the one-level variants is due to the overall small problem size.

5.2. Sedov Problem

The second test case evaluated is a Sedov blast wave [46]. The computational domain consists of one octant of a sphere with radius of 1.2 cm . The initial energy at the origin is chosen such that the shock is located at 1 cm after $1.0\text{ }\mu\text{s}$. For an ideal gas with $\gamma = 5/3$, this corresponds to a total source energy of

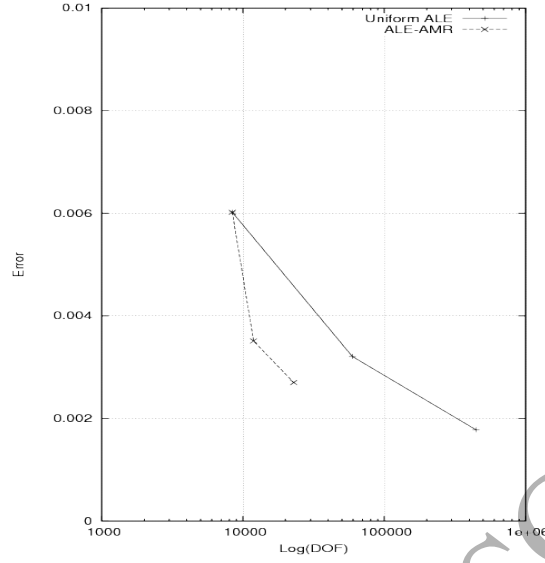


Figure 9: L_1 density error as a function of degrees of freedom for uniform fine mesh ALE and ALE-AMR simulations of the shock tube problem

$0.49359 \text{ g cm}^2/\mu\text{s}^2$. The density field is initialized uniformly to 1 g/cm^3 . Figure 10 shows the computed density contours and the surface mesh for ALE-only, one-level ALE-AMR, and two-level ALE-AMR. Pressure was used as the indicator variable, and the reduction in shock width and overall numerical diffusion due to AMR is clearly evident.

Figure 11 shows density line plots along the diagonal of the volume mesh for the two ALE-AMR cases, along with similar plots for one-level and two-level uniformly refined meshes. The ALE-AMR simulations agree well with the uniformly refined variants, which again indicates convergence of the AMR method toward fine mesh simulation results.

Calculation	L_1 error	Number of cells	Runtime (min)
ALE-only	9.35×10^{-2}	3.6×10^5	2
One-level ALE-AMR	5.91×10^{-2}	1.1×10^6	14
One-level fine mesh ALE	5.78×10^{-2}	2.9×10^6	76
Two-level ALE-AMR	3.34×10^{-2}	3.9×10^6	103
Two-level fine mesh ALE	3.34×10^{-2}	2.3×10^7	790

Table 2: Mesh sizes and runtime information for ALE, ALE-AMR, and uniform fine mesh ALE simulations of the Sedov problem

Table 2 lists the L_1 density error, number of cells, number of time steps, and wall clock time for each variant using 48 parallel threads. For the AMR calculations, the number of cells listed is the value at the end of the simulation, which approximately corresponds to the maximum number due to the nature of the simulation. The convergence rate with AMR is slightly less than first-order, for the same reasons discussed

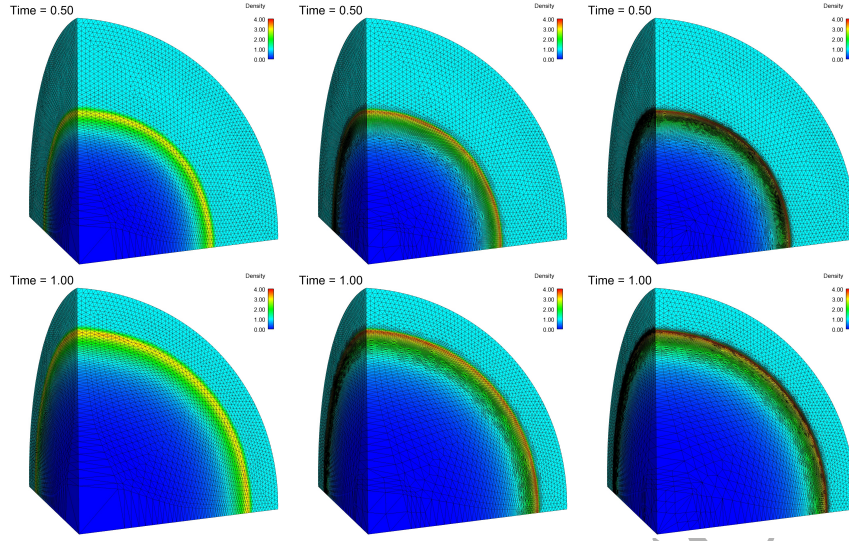


Figure 10: Mesh and density contours for the Sedov problem at intermediate and final times with ALE only (left), one-level ALE-AMR (middle), and two-level ALE-AMR (right)

for the Sod problem. The L_1 density error for the ALE-AMR simulations is within 2 – 3% of the uniform fine mesh ALE simulations for both cases. The use of AMR provides a savings in cell count of approximately $2.6\times$ for the one-level case and $5.9\times$ in the two-level case. This translates to a runtime savings of approximately $5.4\times$ for the one-level case and $7.7\times$ for the two-level case. The error as a function of degrees of freedom is shown in Figure 12, and again shows that ALE-AMR requires fewer degrees of freedom to achieve a given level of error.

5.3. Noh Problem

The third test case evaluated is a Noh problem [47]. The computational domain is a unit cube, initialized to a radial velocity of $-1.0 \text{ cm}/\mu\text{s}$ for all points except the origin, which is a stagnation point. The density is initialized to $1.0 \text{ g}/\text{cm}^3$ and an ideal gas with $\gamma = 5/3$ is used. Figure 13 shows the computed density contours and the surface mesh for ALE-only, one-level ALE-AMR, and two-level ALE-AMR. Density and velocity were both used as indicator variables. AMR reduces both numerical diffusion at the shock front and wall-heating near at origin.

Table 3 lists the L_1 density error, number of cells, number of time steps, and wall clock time for each variant using 16 parallel threads. As before, the number of cells listed for the AMR calculations is the value at the end of the simulation. The convergence rates are slightly less than first-order, but consistent with convergence rates observed for Lagrangian simulations in [44]. The L_1 density error for the ALE-AMR simulations is within 2 – 3% of the uniform fine mesh ALE simulations for both cases. The use of AMR

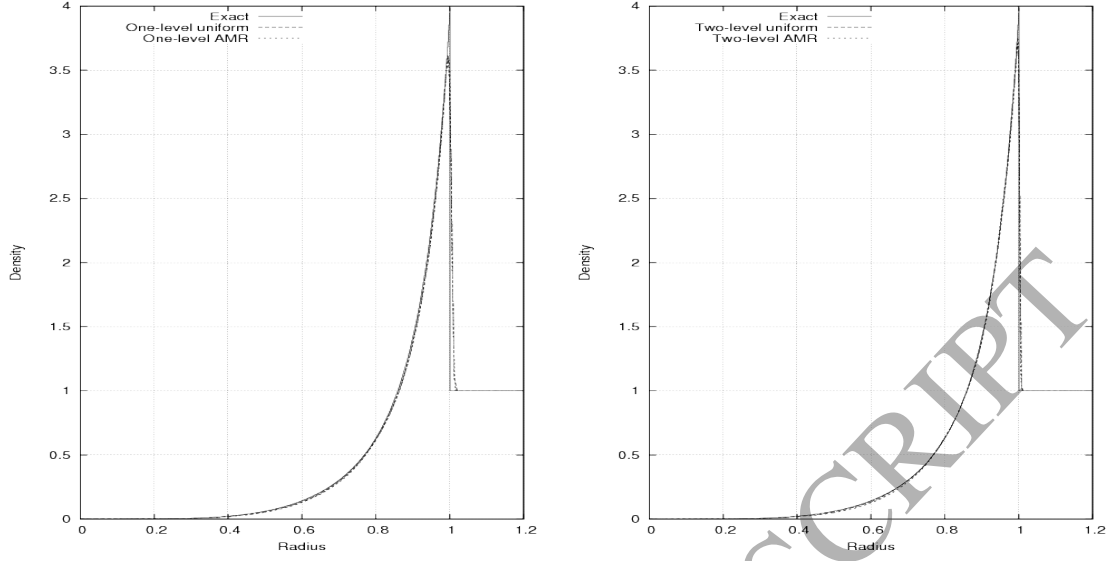


Figure 11: Density line plot for one-level ALE-AMR and one-level fine mesh ALE (left), and two-level ALE-AMR and two-level fine mesh ALE (right) simulations of the Sedov problem, including exact solution

Calculation	L_1 error	Number of cells	Runtime (sec)
ALE-only	1.33×10^0	1.7×10^4	5
One-level ALE-AMR	8.71×10^{-1}	4.0×10^4	15
One-level fine mesh ALE	8.55×10^{-1}	1.6×10^5	45
Two-level ALE-AMR	5.54×10^{-1}	1.9×10^5	114
Two-level fine mesh ALE	5.34×10^{-1}	1.4×10^6	1150

Table 3: Mesh sizes and runtime information for ALE, ALE-AMR, and uniform fine mesh ALE simulations of the Noh problem

provides a savings in cell count of approximately $4.1\times$ for the one-level case and $7.4\times$ in the two-level case. This translates to a runtime savings of approximately $3.0\times$ for the one-level case and $10.1\times$ for the two-level case. The error as a function of degrees of freedom is shown in Figure 15, and again shows that ALE-AMR requires fewer degrees of freedom to achieve a given level of error. As with the Sod problem, the relatively low runtime savings for the one-level variants is due to the overall small problem size.

5.4. Confined Point Blast

Next we reexamine the problem shown in Figure 1, which consists of an idealized blast in a confined geometry. The computational domain is one unit in length, two units in width, and a single cell thick. An ideal gas with $\gamma = 5/3$ and unit initial density is used. The blast is initiated with a single point initialized to a pressure 10^5 times higher than ambient. The solution is shown in Figure 16 at two different times for ALE only and one-level ALE-AMR variants. Density was used as the refinement variable. Comparison of the two calculations shows that the coupled ALE-AMR method successfully refines the coarse mesh regions created

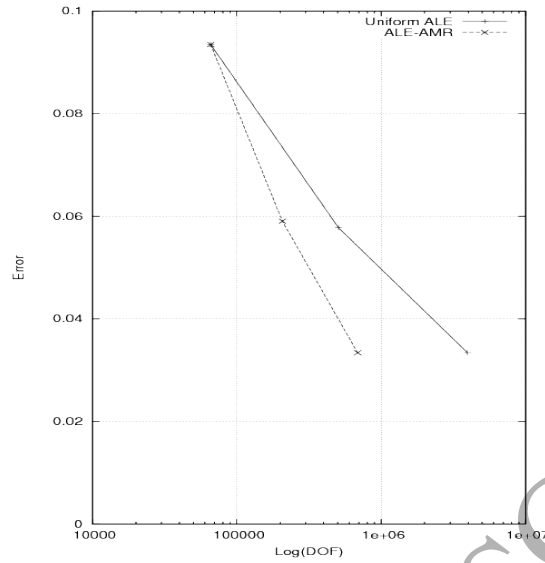


Figure 12: L_1 density error as a function of degrees of freedom for uniform fine mesh ALE and ALE-AMR simulations of the Sedov problem

due to ALE mesh motion away from the blast origin. This test case therefore demonstrates the ability of AMR to mitigate some of the weaknesses associated with ALE methods. Figure 17 shows further detail of the solution at the time of shock impact on the far wall for the two variants. The ALE-AMR variant is clearly less diffusive than the ALE-only simulation

5.5. Triple Point Problem

The last two test cases represent more complicated flows with both shocks and vorticity. The first of these is the triple-point problem [8]. Figure 18 shows the overall dimensions and initial conditions for the 2D planar configuration; a 3D configuration was generated by extruding the 2D configuration by a unit width in the third direction. Since the initial conditions contain jumps in both density and pressure, internal energy was used as the refinement variable.

Figure 19 shows the mesh and density field at two different solution times for ALE-only and one-level ALE-AMR variants. Due to the amount of rotational and translational fluid motion, this problem must run closer to the Eulerian limit than the previous test cases. The unrefined mesh contained approximately 2.0×10^6 cells and the adaptive mesh reached a maximum of approximately 4.3×10^6 cells. This represents a savings of $3.7\times$ over a uniform fine mesh calculation. A close-up of the density field in the vortex region at the final time also is shown in Figure 20. The reduction in numerical diffusion due to AMR relative to the ALE-only calculation is clearly visible.

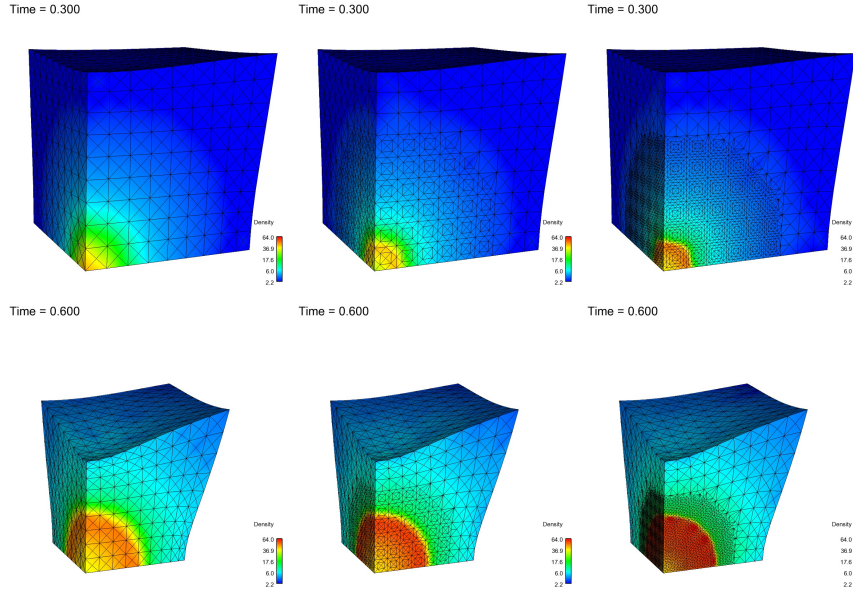


Figure 13: Mesh and density contours for the Noh problem at intermediate and final times with ALE only (left), one-level ALE-AMR (middle), and two-level ALE-AMR (right)

5.6. Shock Interaction with Cylindrical Slug

The final test case involves the interaction of a shock with a high density cylindrical slug and is intended to be more representative of shock hydrodynamics applications. The initial conditions are similar to the previous test problem but with important changes to the overall geometry [48]. First, the high density, low pressure region is truncated at $x = 3$ rather than extending to the boundary at $x = 7$. This geometry change further complicates the shock-vortex behavior. Second, the geometry is rotated into 3D rather than extruded. Figure 21 shows the initial density and pressure distributions for the initial shocked region and the high density slug, as well as on vertical and horizontal cut planes through the volume. This problem also must be run closer to the Eulerian limit.

Figure 22 shows the density field and adapted mesh on a vertical cut plane through the cylindrical geometry at six different simulation times. A comparison of the density field with and without AMR, again on cut planes through the cylindrical geometry, is shown in Figure 23. The various flow features are clearly more sharply resolved with AMR, as expected. The unrefined mesh contained approximately 7.4×10^6 cells and the adaptive mesh reached a maximum of approximately 1.5×10^7 cells. This represents a savings of $3.9\times$ over a uniform fine mesh calculation.

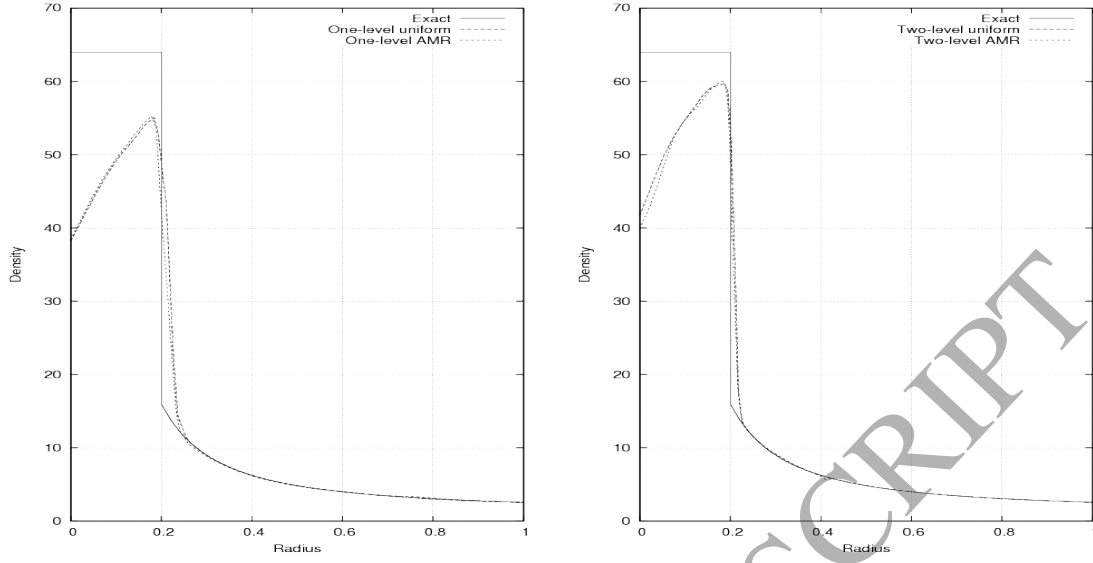


Figure 14: Density line plot for one-level ALE-AMR and one-level fine mesh ALE (left), and two-level ALE-AMR and two-level fine mesh ALE (right) simulations of the Noh problem, including exact solution

6. Conclusion

A coupled ALE-AMR solution method for shock hydrodynamics problems on 3D unstructured grids has been presented. The issues associated with the correct mathematical coupling of ALE and AMR solution algorithms were addressed through both design choices in the individual algorithms, and implicit constraints on the mesh motion of child cells. Convergence of the ALE-AMR solutions towards both uniform fine mesh ALE results and analytical solutions was demonstrated. The benefits of the coupled method were demonstrated on several test cases with meshes of up to $O(10^7)$ tetrahedral cells. Savings in cell count of $3 - 4\times$ were observed for large 3D test problems. The cell count savings translate to speed-ups of $5 - 10\times$ for a given level of error over uniform fine mesh calculations.

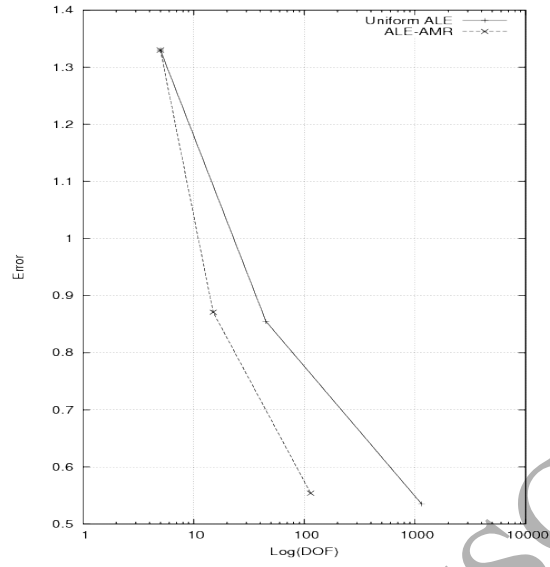


Figure 15: L_1 density error as a function of degrees of freedom for uniform fine mesh ALE and ALE-AMR simulations of the Noh problem

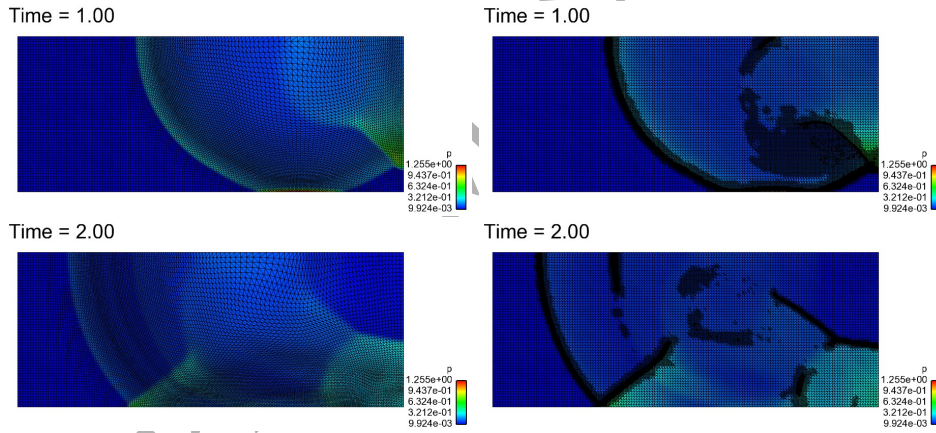


Figure 16: Mesh and pressure contours for a pseudo-2D confined point blast with ALE (left) and one-level ALE-AMR (right)

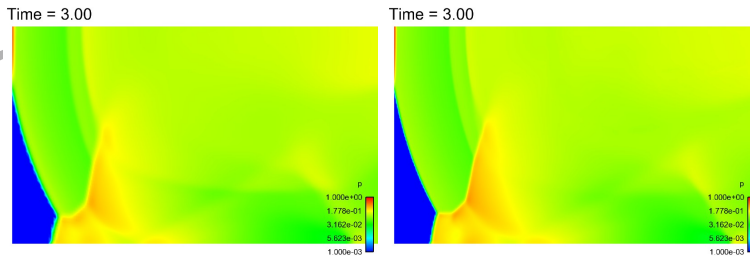


Figure 17: Pressure contours for a pseudo-2D confined point blast with ALE (left) ALE-AMR (right) at the time of shock impact on the far wall

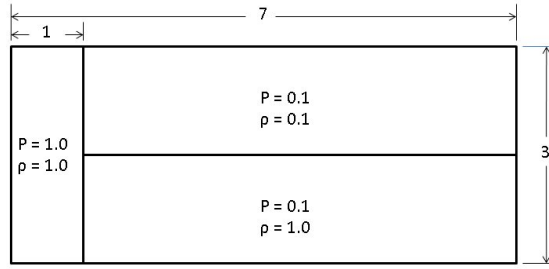


Figure 18: Geometry and initial conditions for the two-dimensional triple point problem

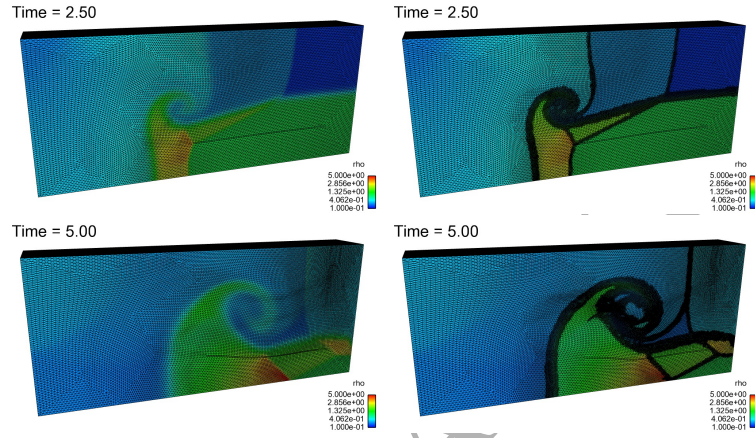


Figure 19: Mesh and density contours for a triple-point problem with ALE (left) and one-level ALE-AMR (right)

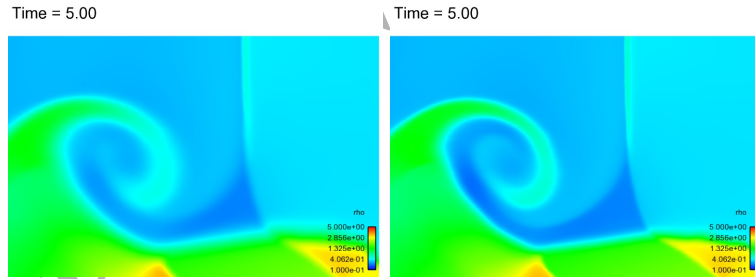


Figure 20: Final density contours for a triple-point problem near the vortex with ALE (left) and one-level ALE-AMR (right)

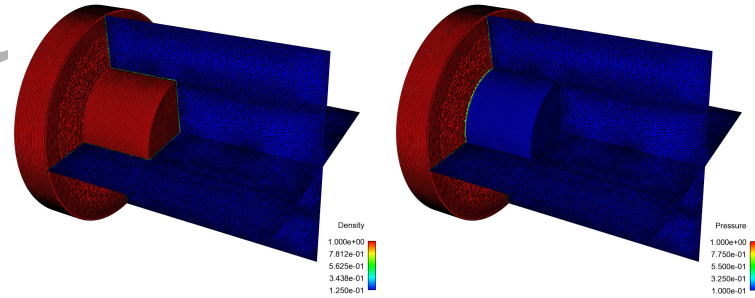


Figure 21: Initial density (left) and pressure (right) distributions and mesh for the shock interaction with a cylindrical slug

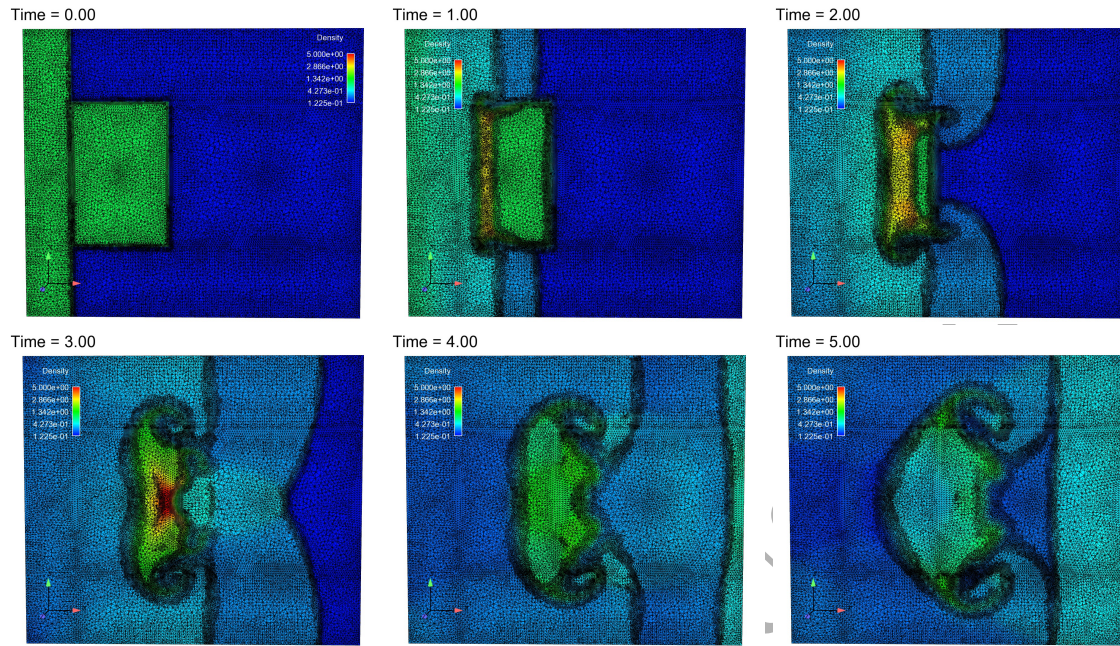


Figure 22: Mesh and density contours for the shock interaction with a cylindrical slug at different simulation times on a vertical cut plane through the cylindrical geometry

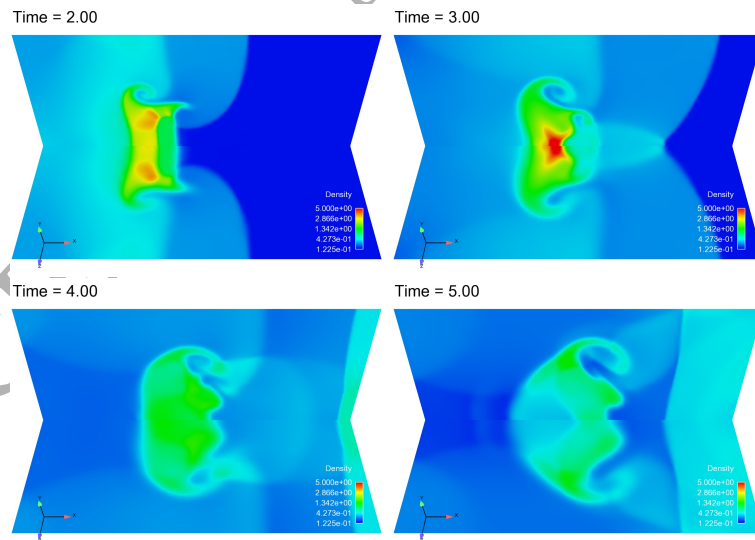


Figure 23: Mesh and density contours for shock interaction with a cylindrical slug at different simulation times with (vertical plane) and without (horizontal plane) AMR

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